BATTERY DISCHARGE ESTIMATION

SPECIFICATION

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DETAILED DESCRIPTION

The magnitude of accumulated entropy generation until complete discharge (AEGD) is applied to rapidly estimate the remaining discharge time (RDT) of lithium-ion (Li-ion) batteries. This approach operates on real-time prediction of RDT during a single discharge cycle and is applicable across diverse operating conditions and battery types. Experimental validation tests are conducted on 18650 and 27000 Li-ion batteries with different capacities and discharge rates. Additional verification test results are presented using independent data from 14500 polymer Li-ion batteries. It is demonstrated that the method accurately identifies RDT for (i) variable operating conditions, (ii) from an arbitrary discharge voltage point, and (iii) fluctuating voltage profiles.

Li-ion batteries are the primary choice for applications ranging from small-scale portable instruments to large-scale electric vehicles. The choice is driven by their favorable characteristics, including high energy and power density, higher cycle life, reduced cost, low self-discharge, etc. Nevertheless, their operational performance and reliability in an application are greatly influenced by the remaining discharge time (RDT), which directly impacts the ability to sustain full discharge.

Accurate prediction of the RDT of lithium-ion batteries plays a critical role in optimizing their operational performance. An effective RDT model can provide valuable insights into the battery's available capacity, preventing issues such as thermal runaway, short circuit fault, overcharging and over-discharging, etc. These factors lead to unexpected shutdowns and potential damage and require substantial expenses for replacement and repairs due to irreversible damage. By safeguarding the battery's integrity through precise capacity forecasting, accurate RDT predictions can significantly extend the lifespan of these essential energy storage systems.

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Various models have been developed to predict the RDT and state of health (SOH) of lithium-ion batteries. Empirical models that curve-fit experimental results are computationally faster; however, they require several empirical constants to fit the full voltage profile and their application is limited to operating conditions tested. The conventional ampere-hour (Ah) approach requires waiting for the entire duration of the battery discharge, while empirical models require information on the depth of discharge (DOD) and necessitate developing equations for different amperage loads. Moreover, they do not consider the physicochemical principles and are restricted to the range of data used in their fitting parameters. Aging and end-of-discharge models incorporate degradation parameters but require detailed aging data and may not capture sudden failures. The Ampere-hour (Ah) method is simple to implement, however, it is highly dependent on the accuracy of the initial state of charge (SOC) and suffers from cumulative errors due to sensor inaccuracies. Machine learning approaches offer fast forecasting but require extensive training data and may be computationally complex for real-time applications. The challenge is developing models that effectively balance complexity, accuracy, and computational efficiency to characterize batteries' discharge behavior.

The premise of this paper is that all systems experiencing degradation during usage are inherently irreversible and dissipative. They generate entropy (d_iS) until equilibrium is reached. According to the second law of irreversible thermodynamics, d_iS for an irreversible dissipative system is always positive and increases monotonically until equilibrium is attained. Building on this principle, researchers have employed the concept of accumulation of entropy generation to track the degradation or aging patterns in brain and cellular functions, tribological wear, coating failures, fatigue-loaded components, electronic components such as resistors, materials undergoing corrosion, etc. Given that the electrochemical energy transformation during the discharge process of Li-ion batteries is inherently irreversible and dissipative, it inevitably results in entropy generating (S_g). This irreversibility stems from internal resistance and other internal or external dissipative factors contributing to entropy generation. To characterize the discharge behavior of batteries and subsequently predict RDT, we propose using the accumulation of the generated entropy to assess full discharge. The entropy generation rate for a Li-ion battery during the discharge process is determined by $\dot{S}_g = \frac{[V_o - V(t)]I}{T(t)}$. Here, I represents the applied discharge current, $V(t)$ and $T(t)$ is the measured discharge voltage and battery temperature in real time, t_d

represents the discharge time to reach 2.5 V, and V_o is the open circuit potential (OCP). The magnitude of accumulation of entropy generation until complete discharge (AEGD), i.e., entropy generation measured from the beginning of the discharge until the voltage reaches to 2.5V (the minimum voltage recommended by the battery supplier), is represented by Eq. (1).

$$AEGD = \int_0^{t_d} \dot{S}_g dt = \int_0^{t_d} \frac{[V_o - V(t)]I}{T(t)} dt \qquad (1)$$

The details of the theoretical modeling for determining $\dot{S}_g$ in equation (1) is discussed associated with later examples.

To establish a procedure for determining the RDT using AEGD, initial tests were conducted on three types of Li-ion batteries under various discharge rates. By analyzing the evolution of entropy generation values, a method was formulated that establishes a correlation between discharge voltage, AEGD, and the state of discharge (SOD) for precise prediction of RDT. AEGD values are computed for battery discharge across two different discharge rates to identify its relationship with the applied current. Using this relationship, a master curve between AEGD and SOD is constructed. To predict RDT in real-time, the measured discharge voltage is used to estimate the corresponding AEGD value, which is then mapped to the SOD equations. By determining the SOD, the procedure calculates the RDT for the battery to reach its cutoff voltage under the given load conditions.

The proposed procedure's effectiveness is demonstrated using commercially available 18650 Li-ion rechargeable batteries with a discharge capacity of 2.5 Ah, tested at discharge rates of 0.25 C, 0.5 C, 1.0 C, and 2.0 C. Extended validation is carried out using published data from independent studies on 14500 Li-ion batteries. The validity of the procedure is further established by predicting the RDT for batteries functioning under variable operating conditions from an arbitrary discharge voltage value and for fluctuating voltage data. We also highlight the practical advantages of the proposed approach over the traditional curve fit method in predicting RDT, emphasizing its ease of implementation and efficiency. Finally, a controller-based setup is developed to verify the real-time prediction of RDT using the proposed procedure during battery discharge.

The proposed procedure for determining the RDT of the battery during discharge and the experimental setup used for testing is described below and in Example Set 1.

The development of a controller-based setup for practical validation of the proposed procedure under various conditions are described in Example Sets 3-5.

The methodology for estimating RDT based on the AEGD value was established by investigating the discharge behavior of three commercially available batteries (Batteries A–C) with varying capacities, each tested at three different discharge rates. The battery type and ratings of the considered batteries are as follows: Battery A (18650, 2800 mAh, 15A), Battery B (21700, 4000 mAh, 35A), and Battery C (18650, 1200 mAh, 30A, Lithium Iron Phosphate ($LiFePO_4$) chemistry). The procedure was subsequently validated using batteries of different capacities, types, and operating conditions.

The experimental procedures, analysis of AEGD evolution, and resulting observations for Batteries A–C is presented in the discussion associated with Example Sets 6-8. The discharge behavior under different operating conditions and AEGD calculations from the measured voltage, current, and battery temperature data, as well as the development of normalization strategies are discussed with the methods employed. These results form the foundation for establishing a generalized relationship between AEGD and the SOD, which is subsequently utilized to predict the RDT of the batteries.

The proposed methodology for estimating the SOD of a battery using AEGD during discharge starts with the calculation of AEGD values for at least two different applied currents or discharge rates. The AEGD is determined using Eq. (1), utilizing measured discharge voltage, contact temperature, and applied current data. Schematic representations of the measured discharge voltages for two applied currents (I_1 and I_2) along with the corresponding evolution trend of AEGD are presented in Figures 42 and 43, respectively. Next, a linear relationship between AEGD and discharge current (I_i) is established from the experimental results, expressed in the form of $AEGD = A(I_i) + B$. A curve-fit equation correlating the evolution of AEGD with SOD at a 1.0 C discharge rate is $SOD = (a \times AEGD^b + c \times AEGD)$, where *a*, *b*, and *c* are the curve-fit coefficients. To generalize the model across various discharge rates, a ratio (R_x) is introduced. R_x is defined as the AEGD value at 1.0 C divided by the AEGD value at other discharge rates, i.e., $R_x = \frac{AEGD_{1.0}}{AEGD_x}$. Using this ratio, a generalized equation relating SOD to AEGD for all discharge rates is formulated as shown in Eq. (5):

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$$SOD_x = (a \times AEGD^b + c \times AEGD) \times R_x \tag{5}$$

The schematic representation of AEGD versus SOD is shown in Figure 44. The RDT is subsequently calculated using the predicted SOD, the battery's rated capacity (Ah), and the applied current (*I*), according to Eq. (6):

$$RDT = \frac{Ah}{I} \times 3600 \times \frac{(100 - SOD)}{100} \tag{6}$$

Example Set 1 - Constant discharge conditions

Figure 45 shows the test setup employed to validate the proposed approach. This setup measures the discharge voltage for the applied current and determines the entropy generation during a single battery discharge. The setup consists of a programmable battery tester, a thermocouple, and a data acquisition system (DAQ) to measure the temperature data from the thermocouple. The battery's discharge rate is controlled by programming the battery tester via its software. The tester can provide a constant current to the Li-ion battery and measure the output voltage. A maximum current of 5A can be applied with an accuracy of 0.01 A. The resolution of the voltage measurement is 0.01 V, and the minimum cutoff voltage is 0.05 V. The voltage and current data from the tester are recorded in real-time at one data set per second. All batteries are fully charged prior to conducting the discharge test. Once fully charged, the batteries are subjected to several constant, variable, and fluctuating amperage loads until the voltage reaches below 2.5 V, as per the supplier's recommendation.

Battery D, a 2.5 Ah 18650 Li-ion battery, was tested under constant discharge conditions. The effectiveness of the proposed procedure is evaluated by predicting the RDT of Battery D under four different constant discharged rates. These discharge rates—0.25, 0.5, 1.0, and 2.0 C—are obtained by applying currents of 0.63, 1.25, 2.5, and 5.0 A, respectively. The discharge voltage measured for all four discharge rates is provided in Figure 1. To predict the RDT using a linear curve fit equation correlating the AEGD value with the corresponding discharge currents, employing at least two discharge data are needed. For this purpose, the data for 0.5 and 1.0 C discharge rates and the RDT for the remaining discharge rates are utilized.

The AEGD values for 0.5 and 1.0 C discharge rates are calculated using Eq. (1) and the evolution of AEGD is plotted in Figure 2. The linear relationship between the AEGD value at 2.5V and the corresponding discharge current is: $AEGD = 0.577I + 10.2$. Subsequently, for the

1.0 C discharge rate, the AEGD value is estimated with respect to the SOD (see Figure 3), and the equation correlating AEGD to SOD is: $SOD = (185.7 \times AEGD^{0.99} + 175.6 \times AEGD)$. A generalized equation for estimating the SOD across all discharge rates is provided in Eq. (7) and in this equation R_x is calculated as the ratio of the AEGD value at a given discharge rate to the AEGD value at 1.0 C.

$$SOD_x = (185.7 \times AEGD^{0.99} + 175.6 \times AEGD) \times R_x \qquad (7)$$

Thus,

$$RDT_x = \frac{Ah}{I} \times 3600 \times \frac{(100 - SOD_x)}{100}.$$

The predicted and measured evolution of RDT for all four discharge rates is shown in Figure 4. This figure shows that the predicted RDT closely follows the measured RDT, with errors of 0.4, 0.94, 1.1, and 4.4% at 2.5 V for discharge rates from 0.5 C to 2.0 C.

Figures 1-4 relate to the demonstration of the proposed procedure for Battery D: 18650 Li-ion battery with 2.5 Ah. Figure 1 depicts discharge voltage measured for 0.25, 0.5, 1.0, and 2.0 C discharge rates, Figure 2 depicts evolution of AEGD for 0.5 and 1.0 discharge rates. Figure 3 depicts percentage SOD versus AEGD for 1.0 C discharge rates, for developing a curve-fit equation, Figure 4 depicts measured and predicted RDT for all discharge rates.

To further validate the proposed procedure, we utilized real-time voltage data for a 14500 Li-ion battery with lithium cobalt oxide ($LiCoO_2$) cathode material, as reported in an independent published study. Figure 5 shows the voltage corresponding to various discharge rates (0.5, 1.0, and 2.0 C). As the discharge current magnitude is not specified, it is assumed that the discharge rate represents the current.

To predict the RDT in the present case using the proposed procedure, data from the 1.0 and 2.0 C discharge rates are used to establish a linear correlation between the AEGD value and the discharge rate. The RDT is subsequently determined for the 0.5 C discharge rate. The AEGD value for 1.0 and 2.0 C discharge rates is calculated using Eq. (1), and the linear relationship between the AEGD values at 2.5 V and discharge rates is formulated as $AEGD = 1.72I + 4.88$. Utilizing this equation, the AEGD value for the 0.5 C discharge rate is 5.7. Furthermore, referring to Figure 6, the evolution of AEGD with SOD for the 1.0 C discharge rate is: $SOD = (629 \times AEGD^{0.99} + 1602 \times AEGD)$. The generalized SOD for all discharge rates is provided in

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Eq. (8). The ratios of the AEGD values for the 0.5 and 2.0 C discharge rates to the AEGD value at the 1.0 C discharge rate are calculated as $R_{0.5}$ = 1.168 and $R_{2.0}$ = 0.79, respectively.

$$SOD_x = (629 \times AEGD^{0.99} + 1602 \times AEGD) \times R_x \quad (8)$$

Using the identified linear curve-fit between AEGD and discharge rate, along with equation (8), the RDT is calculated and plotted in Figure 7 alongside the experimentally measured RDT values. Across all three discharge rates, the maximum error between the measured and predicted RDT values is approximately 5%. Thus, the proposed procedure is capable of providing real-time prediction of the RDT.

Figures 5-7 depict the validation of the proposed procedure for the 14500 Li-ion battery. Figure 5 depicts evolution of discharge voltage for 0.5, 1.0 and 2.0 discharge rates. Figure 6 depicts the percentage of SOD variation for the evolution of AEGD for the 1.0 C discharge rate. Figure 7 depicts measured and predicted RDT for all discharge rates.

Example Set 2 - Variable operating conditions

Batteries often experience dynamic loading conditions rather than constant discharge rates. Therefore, validating predictive models for batteries operating under variable discharge rates is essential for ensuring their reliability in real-world conditions. The proposed procedure's efficacy under two variable discharge conditions for Batteries A and B is examined below. The batteries are tested by ramping down and up at three discharge rate levels. The selected variable conditions selected for these tests are detailed in Table 1.

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Table 1 Duration of application of variable current to Batteries A and B

Battery	Variable Loads	Current	Duration (s)
Battery A	Ramp down	1.25	2400
		2.5	1200
		5.0	Until complete discharge
	Ramp up	5.0	900
		2.5	1200
		1.25	Until complete discharge
Battery B	Ramp up	1.25	3600
		2.5	1800
		5.0	Until complete discharge
	Ramp down	5.0	900
		2.5	1800
		1.25	Until complete discharge

Figure 8 illustrates the discharge voltage of Battery A under varying discharge currents applied in Case 1 (ramping up: 1.25 → 2.5 → 5.0 A) and Case 2 (ramping down: 5.0 → 2.5 → 1.25 A). The AEGD is calculated using Eq. (1). Considering, $R_{0.5} = 1.05$, $R_{2.0} = 0.9$, and the equation for SOD (Eq. (A11)), the evolution of SOD with AEGD is determined and shown in Figure 9. The AEGD evolution at a 1.0 C discharge rate is also included for reference. Comparing Figures 9 and 46, although the AEGD evolution over time differs, the proposed procedure ensures that the trends remain consistent under both constant and variable operating conditions. Figure 10 shows the predicted RDT for both cases over time. The actual time for the battery to reach 2.5 V was measured as 4597 s for Case 1 and 4903 s for Case 2. The predicted error between the measured and calculated RDT at 2.5 V for Cases 1 and 2 is 0.1% and 0.4%, respectively.

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To assess Battery B's response to the varying discharge current, the battery was tested at two different cases: Case 1, where the discharge current was ramped up (1.25 → 2.5 → 5.0 A), and Case 2, where the current was ramped down (5.0 → 2.5 → 1.25 A). Figures 11 and 47 show the measured discharge voltage and calculated AEGD behavior. Figure 12 illustrates the evolution of SOD with AEGD, confirming that the proposed procedure ensures consistency in their behavior under both constant and variable operating conditions. Figure 13 presents the predicted RDT for both cases until the battery reaches 2.5 V, with measured discharge times of 6429 s for Case 1 and 6657 s for Case 2. The predicted error between the measured and predicted time at 2.5 V discharge voltage is 2.1% and 3 % for Cases 1 and 2, respectively.

Based on the discussion above, the predicted RDT values obtained through the proposed procedure are closely aligned with the measured values for Batteries A and B. This indicates that the method effectively provides accurate RDT predictions under variable loading conditions.

Figures 8-11 relate to the validation of the proposed procedure under variable operating conditions. Figures 8-10 show measured discharge voltage under different applied currents, the relationship between AEGD and the percentage SOD, and predicted RDT for Cases 1 and 2 for Battery A. Figures 11-13 show measured discharge voltage, AEGD versus SOD plot, and predicted RDT for Cases 1 and 2 for Battery B.

Example Set 3 - Predicting RDT from Intermediate Discharge Voltage

In the preceding discussion, we outlined the proposed procedure for determining the RDT, specifically for batteries after reaching a full charge or at the start of the discharge cycle. However, in real-world applications, RDT estimation is often required at an intermediate discharge voltage rather than from the start of discharge. To address this, we extend the proposed methodology to determine RDT between 10% and 90% SOD, to ensure its applicability as an adaptable prediction framework. To achieve this, we propose establishing a relationship between the discharge voltage and the AEGD value, allowing us to predict RDT from any arbitrary voltage.

A battery's discharge voltage curve typically exhibits three distinct phases: an initial transient phase, a linear operating region, and a final transient decline (see Figure 48). Since the linear region accounts for approximately 80% of the battery's operational lifespan, we propose selecting arbitrary voltage points within this region to improve the accuracy and reliability of

RDT predictions. This approach ensures that the estimated discharge time is both representative of real-world conditions and effectively captures the battery's degradation behavior over time.

To establish the procedure for determining the RDT from an intermediate discharge voltage, we analyze the variation of AEGD with discharge voltage between 10% and 90% of the SOD for Batteries A, B, and D. Figures 14-16 illustrate the relationship between discharge voltage and AEGD for Battery A at 0.5, 1.0, and 2.0 C, Battery B at 0.31, 0.63, and 1.25 C, and Battery D at 0.25, 0.5, and 1.0 C discharge rates. The results indicate that the AEGD variation with discharge voltage follows a consistent trend across different discharge rates and has an approximately linear variation. Utilizing this linear relationship, AEGD can be estimated for any intermediate discharge voltage, allowing for the subsequent prediction of RDT using the proposed methodology.

To validate, let us consider an arbitrary discharge voltage of 3.6 V and determine the RDT for all considered batteries. At this voltage, the AEGD values calculated for Batteries A, B, and D are 9.17 J/K, 11.37 J/K, and 7.12 J/K, respectively. Using these AEGD values, the SOD for Batteries A, B, and D is determined across all discharge rates by applying Eqs. (A11), (A12), and (A13), respectively. The measured and predicted SOD values are presented in Figure 17, showing that the error between measured and predicted SOD ranges from 0.8% to 4.2%.

These results confirm that the proposed methodology effectively translates AEGD and discharge voltage relationships into reliable RDT predictions. By leveraging this approach, RDT can be accurately determined from any arbitrary voltage within the battery's operating range rather than requiring data from the beginning of the discharge cycle. This capability is particularly beneficial for real-time battery management systems, where predicting remaining battery life from a given voltage is critical for optimizing energy utilization, ensuring system reliability, and preventing unexpected power losses.

Figures 14-17 validate the proposed procedure using intermediate discharge voltage. Figures 14-16 depict measured discharge voltage profiles for 10% to 90% SOD with corresponding AEGD values for Batteries A, B, and D, respectively. A linear-fit trendline is included in each plot to illustrate the relationship between discharge voltage and AEGD, demonstrating a consistent and predictable pattern that supports accurate RDT estimation from arbitrary voltage points. Figure 17 shows the comparison of measured and predicted SOD for

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Batteries A, B, and D at various discharge rates with experimental results appearing in the left bar of each pair of results and predicted values appearing in the right bar of each pair of results.

Example Set 4

Next, we apply the proposed procedure to fluctuating discharge voltage data, which commonly occurs due to variations in load conditions, internal resistance, and battery aging effects. To accomplish this, we examine a dataset that includes fluctuating voltage readings alongside a non-varying discharge voltage profile for the tested battery (Figure 49). Using this dataset, we compute the AEGD value by applying Eq. (1). The resulting evolution of the AEGD value is shown in Figure 50. From this figure, it can be concluded that despite the presence of fluctuating data, the evolution of the AEGD value remains unaffected. This is attributed to the cumulative nature of AEGD, which tends to smooth out transient variations and ensures consistency in its progression. This observation demonstrates the robustness of the proposed method in accurately estimating AEGD and its applicability under dynamic operating conditions.

Example Set 5 - Comparison with Curve-fit Approach - an Empirical Model

Several empirical models are used to characterize the single discharge behavior of Li-ion batteries and estimate the RDT. The Single Particle Model (SPM) provides an analytical approach for modeling solid-phase diffusion in electrode particles and requires parameters such as the solid diffusion coefficient, exchange current density, electrode thickness, and conductivity to predict RDT accurately. The Doyle-Fuller-Newman (DFN) model offers a more comprehensive approach by incorporating both solid and electrolyte dynamics. While highly accurate, it is computationally intensive and requires parameters including solid and electrolyte diffusion coefficients, electrode and electrolyte conductivity, reaction rate constant, and transference number for RDT estimation. In contrast, the Equivalent Circuit Model (ECM) is a simplified representation of the battery using resistors, charge transfer resistance, and double-layer capacitance. While suitable for real-time applications, it is less precise in predicting degradation effects. Since these analytical approaches rely on detailed battery material parameters, and this study focuses on commercial batteries with limited parameter availability, we test the effectiveness of the curve-fitting approach.

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The present method is evaluated by comparing the prediction error of RDT using a curve-fitting approach for the discharge curves of battery D, provided in Figure 1, and in Example Set 1. The empirical approach will be validated by developing a curve-fit equation for the discharge voltage as a function of SOD using 0.5 and 1.0 C discharge rates and extended for 0.25 and 2.0 C discharge rates. A fifth-degree polynomial curve fit equation provided in Eq. (9) will be applied for 0.5 and 1.0 C discharge rates data and the six coefficients ($p_0,\ p_1,\ p_2,\ p_3,\ p_4,$ and p_5) will be determined. The resulting coefficients ($p_0 - p_5$) are linearly extrapolated to predict coefficients for other discharge rates (0.25 and 2.0 C). Using these coefficients, a fifth-degree polynomial curve is constructed to estimate the SOD based on the measured discharge voltage. The discharge voltage with respect to the percentage of SOD is shown in Figure 18.

$$SOD = p_0 + p_1 V + p_2 V^2 + p_3 V^3 + p_4 V^4 + p_5 V^5 \qquad (9)$$

Figure 19 presents the SOD percentage along with the discharge curves for 0.5 and 1.0 C discharge rates, as well as the predicted SOD for the measured voltage using Eq. (9). The obtained coefficients ($p_0 - p_5$) are linearly fitted against the applied discharge rates to determine the coefficients for other discharge rates (0.25 and 2.0 C). Using these coefficients, the predicted SOD percentage for the measured discharge voltages of 0.25 and 2.0 C discharge rates are shown in Figure 20. The error percentage between the measured and predicted SOD is depicted in Figure 21.

Figure 18 depicts measured discharge voltage with % SOD for the different amperage loads of 1.5, 2.0, 2.5, and 3.0 A, for Battery D. Figure 19 shows an overlay plot between the measured and calculated discharge voltage from the polynomial curve-fit equation for 0.5 and 1.0 C discharge rates. Figure 20 shows extended prediction of the discharge voltage for discharge current of 0.25 and 2.0 C, using a fifth order polynomial fit. Figure 21 shows the measured percentage error with % SOD.

Thus far, the proposed procedure was validated using a battery tester, relying primarily on recorded data rather than real-time measurements. While this approach provided insights into the procedure’s effectiveness, it limits the analysis to post-data evaluations of the recorded data. To enhance the validation process and ensure real-time applicability, a comprehensive setup integrating a controller-based measurement system is designed and tested (see Figure 22). This system allows for continuous monitoring of key battery performance parameters, such as voltage,

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current, and temperature during the discharge process. In the present study, the real-time analysis is achieved using a microcontroller.

In this real-time validation setup, the battery is subjected to a controlled load via the battery tester, allowing it to simulate desired discharge conditions. The controller system will capture real-time data as the battery discharges, facilitating instantaneous RDT analysis. This integration is critical for assessing the effectiveness of the proposed procedure under actual operating conditions, as it enables the evaluation of the RDT with real-time data. In this setup, the discharge voltage is directly acquired in real-time using the controller’s analog input, while a current sensor (ACS712 (Hall-effect based) with 0-5A range) is used to measure the applied discharge current by the battery tested and a K-type thermocouple with MAX6675 is used to monitor temperature variations during operation. Following the assembly and calibration of the circuit, experiments were performed on the battery at various discharge currents (1.25A, 2.5A, and 5A) to assess the accuracy and reliability of the developed procedure. The evolution of the real-time measured discharge voltage and SOD for 1.25, 2.5, and 5A discharge currents are presented in Figures 23-25, respectively. On these figures, the predicted discharge times at the conclusion of the experiment or after reaching 2.5 V are shown. The predicted discharge durations for 1.25, 2.5, and 5A were found to be 51, 20, and 11 s, respectively.

Figures 22-25 show the real-time experimental validation of the proposed procedure using a micro-controller. Figure 22 shows an experimental setup to measure the real-time discharge voltage, applied current, and temperature using a micro-controller. Figures 23-25 show real-time measured discharge voltage and SOD, along with predicted RDT for applied current of 1.25, 2.5, and 5.0 A discharge current is shown.

The results obtained across different applied currents showed a high correlation with expected values, demonstrating the effectiveness of the setup in capturing real-time battery behavior. The system successfully provided reliable data for AEGD-based predictions, reinforcing the potential for using this method in practical applications such as battery health monitoring and performance analysis. Implementing this controller-based measurement system provides a cost-effective and scalable solution for further research in battery diagnostics and real-time monitoring.

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This study aims to develop a simplified thermodynamic framework for accurately predicting the RDT of lithium-ion rechargeable batteries using the concept of AEGD. The results confirm that this approach not only improves RDT prediction accuracy but also provides substantial practical benefits over conventional analytical methods, particularly in real-time applications.

The effectiveness of the proposed methodology was validated through experimental testing on Battery D, which was discharged at four different discharge rates: 0.25, 0.5, 1.0, and 2.0 C. The AEGD-based predictions demonstrated strong agreement with the measured RDT values, highlighting the robustness of this approach across varying discharge conditions. The observed prediction error, ranging from 0.4% to 4.4%, underscores the high level of precision achievable with this technique.

Conventional RDT prediction methods often rely on empirical data and curve-fitting techniques constrained by the specific conditions under which they were developed or use machine learning, requiring large datasets to make predictions. In contrast, the AEGD-based framework leverages fundamental thermodynamic principles, enabling a more generalized and adaptable solution. This versatility has been demonstrated through extensive validation with 18650, 27000, and 14500 battery cells, confirming its applicability across different battery chemistries and operating conditions. Furthermore, the model's ability to maintain accuracy under fluctuating current profiles reinforces its reliability for practical applications.

Integrating a controller-based real-time monitoring system further enhances the practical implications of this research. The system dynamically recalibrates RDT predictions based on real-time battery performance data by continuously tracking discharge current, voltage, and surface temperature. This capability not only streamlines the validation process but also contributes to the advancement of intelligent battery management systems, optimizing both performance and safety during discharge cycles.

This study presents a simplified thermodynamic framework utilizing AEGD to enable real-time prediction of the RDT of lithium-ion batteries. The proposed methodology establishes a generalized relationship between SOD and applied current, achieving accurate RDT predictions across varying battery types, discharge rates, and different types of operating (constant and variable) conditions. Experimental validation, including comparisons with published data and fluctuating load scenarios, establishes the robustness of the proposed method, with prediction errors of less than 5%. Compared to curve-fitting empirical models that require empirical adjustments, the AEGD-based approach offers a generalized thermodynamic-driven solution that can adapt to different battery chemistries and discharge profiles. Furthermore, the successful integration of a controller-based real-time monitoring system highlights the practicality of this methodology for real-world applications, paving the way for enhanced battery management systems that improve safety, reliability, and operational efficiency.

The descriptions below provide a detailed theoretical framework to derive the equation for calculating entropy generation during the discharge of Li-ion batteries. This formulation is based on integrating fundamental principles, including the first and second laws of thermodynamics, followed by the concepts from Gibbs free energy.

According to the first law of thermodynamics, the energy balance equation for a closed battery system undergoing discharge is expressed as:

$$dU = dW + dQ + \sum_k \eta_k dN_k \tag{A1}$$

where dU is the change in internal energy, dW is the change in work done with the application of an external load, and dQ is heat dissipated across the system boundary. The chemical potentials η_k and number of moles N_k of species k, signifies the contribution from the electrochemical process, with no mass flow across the system boundary. Parameters η_k and N_k are chemical potentials and the number of moles of species k, respectively, with no mass flow across the battery boundary. The total change in energy $\sum_k \eta_k dN_k$ is composed of electrochemical kinetics due to chemical reactions $\sum_k \eta_k^r dN_k^r$ and ionic diffusion $\sum_k (\eta_{high} - \eta_{low}) dN_k^d$ (see Eq. (A2)).

$$\sum_k \eta_k dN_k = \sum_k \eta_k^r dN_k^r + \sum_k (\eta_{high} - \eta_{low}) dN_k^d \tag{A2}$$

where η_{high} and η_{low} are chemical potentials in the high- and low-potential regions, respectively and η_k^r is the chemical potential during the chemical reaction. It has been demonstrated that theses chemical reaction prevailing in the battery leads to the generation of electrical energy in the battery. Therefore, in Eq. (A2), the right-hand side term is replaced with directly coupled electrical boundary work Vdq. Eq. (A2) evolved to Eq. (A3).

$$\sum_k \eta_k dN_k = Vdq \tag{A3}$$

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By substituting the rate of charge transfer $\dot{q} = \left(\frac{dq}{dt}\right)$ equal to the current (*I*), considering the operation of the batteries in the atmospheric pressure condition ($\text{i.e.}, dP = 0$) and that the battery does not experience a change in its volume, the equation of internal energy rate is determined as provided in Eq. (A4).

$$\dot{U} = \dot{Q} + VI \tag{A4}$$

According to the second law of thermodynamics, the entropy is not conserved and increases during the battery discharge process. The change in entropy of the system, denoted as dS, consists of irreversible entropy generation dS_g and reversible entropy flow dS_e:

$$dS = dS_g + dS_e \tag{A5}$$

The entropy flow arises from heat transfer via heat flow dQ to the surrounding and substituting $\dot{S}_e = \dot{Q}/T$ in Eq. (S5) and substituting $\dot{S}$ in Eq. (A4), the rate of internal energy is given by:

$$\dot{U} = T\dot{S} - T\dot{S}_g + VI \tag{A6}$$

Gibbs free energy, G, offers a framework for evaluating the operational conditions of batteries under atmospheric pressure without volume change. The fundamental relationship governing Gibbs free energy is expressed as:

$$\dot{G} = \dot{U} - T\dot{S} - S\dot{T} \tag{A7}$$

Substituting Eq. (A7) for $\dot{U}$ in Eq. (A6), gives the Gibbs fundamental relation for the battery (see Eq. (A8)).

$$\dot{G} = VI - S\dot{T} - T\dot{S}_g \tag{A8}$$

We can evaluate the Gibbs free energy based on the potential during open-circuit V_o multiplied by the charge transfer over time *dq* (*Idt*). This product represents the energy dissipated internally during the discharge process, acknowledged as Electro-Chemical-Thermal (ECT) energy, which is $S\dot{T} = -C\dot{V}$. Substituting $\dot{G} = V_o I$ and $S\dot{T} = -C\dot{V}$ in Eq. (A8) and solving for $\dot{S}_g$ yields Eq. (A9).

$$\dot{S}_g = \frac{VI - V_o I}{T} + \frac{C\dot{V}}{T} \tag{A9}$$

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Research has shown that the magnitude of the contribution of the second term on the right-hand side of Eq. (A9) is negligible and therefore, the second term is neglected. The final equation for the entropy generation is Eq. (A10)

$$\dot{S}_g = \frac{(V_o - V)I}{T} \tag{A10}$$

where V_o is the open-circuit voltage (OCP), represents the voltage of a battery when it is not connected to a load. The OCP of a battery typically depends on the state of charge (SOC) and is specific to its chemistry. However, various factors often complicate deriving an accurate OCP-SOC relationship, including temperature variations, charging rates, and the inherent hysteresis effects observed in battery systems. OCP can be determined using two primary methods, each differing in complexity and computational requirements. The first method consists of measuring the voltage across a battery's terminals using a voltmeter or DAQ after disconnecting the load and allowing the battery to rest for a sufficient period, ensuring transient effects dissipate and internal equilibrium is restored. For precise results, it is crucial to minimize external current drawing, as even minor currents can impact the readings. Factors such as temperature fluctuations and battery chemistry should also be considered to enhance measurement accuracy. The second approach for determining OCP is to use a close-to-equilibrium OCV measurement approach. This involves cycling a reference battery cell at a very low C/20 or C/25 rate, which minimizes polarization effects and allows the practical capacity to be close to the maximum attainable and fitting SOC of the battery. The pseudo-OCV is then derived as the mean voltage from both the charge and discharge curves at this low rate. The limitation of this method is that it provides only an approximate estimation of the OCV value, and testing at a C/25 rate considerably extends the measurement duration. Moreover, obtaining a precise OCV prediction requires testing at rates lower than C/25.

To streamline the OCP calculation process and enhance computational efficiency, a common approach is to assume OCP as a constant value and a constant value of 4.3 V is adopted in the present work. Given this simplification, we refer to the proposed method as a simplified thermodynamic approach, as it leverages fundamental thermodynamic principles while eliminating computationally associated with treating V_o as variable and intensive elements typically required for more detailed electrochemical modeling. Although this approach may introduce minor inaccuracies compared to more complex models, it offers practical advantages in scenarios where computational efficiency, real-time estimation, or resource constraints are key considerations.

Example Set 6

The discharge voltage and battery surface temperature plots of Battery A, a 18650 Li-ion battery with 2.8 Ah, in its pristine condition at different discharge rates (0.5, 1.0, and 2.0 C) is shown in Figures 26 and 27, respectively. The discharge rates of 0.5, 1.0, and 2.0 C are achieved by applying 1.25, 2.5, and 5 A currents to the battery. Using the measured discharge voltage, contact temperature, and applied current data, the evolution of the AEGD is calculated using Eq. (1) and presented in Figure 28. The AEGD value at a discharge voltage of 2.5 V is shown in Figure 29. From this figure, a linear relationship between the AEGD and applied current is observed, and the linear curve-fit equation connecting them is determined to be $AEGD = 1.07I + 22.61$. The evolution of the AEGD with SOD is determined and provided in Figure 30. From this figure, it is inferred that the evolution of AEGD with SOD varies with current.

Subsequently, taking into consideration that the final AEGD value at 2.5 V discharge voltage exhibits a linear relationship with the current, the progression of AEGD with SOD for 0.5 and 2.0 C discharge rates is normalized in reference to the 1.0 C discharge rate. This normalization is achieved by calculating the ratio of AEGD at 1.0 C at a 2.5 V discharge voltage to the AEGD values at the other discharge rates, which is then multiplied by the corresponding AEGD evolution values. For example, the ratios of AEGD at 1.0 C compared to 0.5 and 2.0 C discharge rates are $R_{0.5} = \frac{AEGD_{1.0}}{AEGD_{0.5}} = 1.05$ and $R_{2.0} = \frac{AEGD_{1.0}}{AEGD_{2.0}} = 0.9$, respectively. To normalize the evolution of AEGD values corresponding to 0.5 and 2.0 C, discharge rates are performed by multiplying the corresponding *R* values. The normalized AEGD values with time are plotted against SOD in Figure 31, alongside the data for the 1.0 C discharge rate. This figure demonstrates that the evolution of AEGD with SOD at different currents closely follows the same trajectory. In other words, by multiplying the identified ratios of $R_{0.5}$ and $R_{2.0}$ to the curve-fitting equation for AEGD and SOD at the 1.0 C discharge rate will pave a pathway to determine the progression of AEGD with SOD for other discharge rates.

The curve-fit equation identified for 1.0 C discharge rate with SOD is given by $SOD = 219.6 \times AEGD^{0.99} - 208.6 \times AEGD$. The generalized curve-fitting equation for all discharge rates, incorporating the ratios (R_x), is given in Eq. (A11), where x represents the discharge rate. Employing Eq. (A11), from the AEGD value, the state of discharge can be identified, followed by predicting the remaining useful life.

$$SOD_x = (219.6 \times AEGD^{0.99} - 208.6 \times AEGD) \times R_x \qquad \text{(A11)}$$

Figures 26-31 show experimental results for Battery A: 18650 Li-ion battery with 2800mAh. Figure 26 shows discharge voltage measured for different discharge current rates (0.5, 1.0, and 2.0 C). Figure 27 shows measured battery temperature rise for 0.5, 1.0, and 2.0 C discharge rates. Figure 28 shows evolution of AEGD for different discharge rates. Figure 29 shows AEGD at 2.5 V with different discharge currents and its linear fit. Figure 30 shows the evolution of AEGD for different discharge rates with percentage of SOD. Figure 31 shows the normalized evolution of AEGD for different discharge rates with percentage of SOD.

Example Set 7

The discharge voltages measured for Battery B, a 21700 Li-ion battery with 4.0 Ah, in the pristine state at various discharge rates (0.31, 0.63, and 1.25 C) are provided in Figure 32. These discharge rates are achieved by applying 1.25, 2.5, and 5 A current, respectively. The lower discharge rate is a result of the battery tester's limitation, which restricts the maximum applied current to 5 A. Using Eq. (1), the evolution of AEGD is determined and presented in Figure 33. Figure 34 presents the AEGD value up to 2.5V, reaffirming the linear relationship between AEGD and I, expressed as $AEGD = 0.81I + 28.7$. The evolution of AEGD with SOD is displayed in Figure 35. Subsequently, the AEGD evolution for 0.31 and 1.25 C discharge rates is normalized relative to the 0.63 C discharge rate using the calculated ratios $R_{0.31} = 1.032$, $R_{1.25} = 0.937$ and the results are plotted in Figure 36. The plot shows that AEGD evolution with SOD follows a consistent pattern across different currents. A generalized curve-fitting equation incorporating discharge rate ratios is presented in Eq. (A12), enabling SOD determination from AEGD values.

$$SOD_x = (193 \times AEGD^{0.99} - 183.2 \times AEGD) \times R_x \qquad \text{(A12)}$$

Figures 32-36 show experimental results for Battery B: 21700 Li-ion battery with 4000 mAh. Figure 32 shows discharge voltage measured for different discharge current rates (0.31, 0.63, and 1.25 C). Figure 33 shows evolution of AEGD for different discharge rates, Figure 34

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shows AEGD at 2.5 V with different discharge currents and its linear fit. Figure 35 shows evolution of AEGD for different discharge rates with percentage of SOD. Figure 36 shows normalized evolution of AEGD for different discharge rates with percentage of SOD.

Example Set 8

Figure 37 presents the discharge voltages measured for Battery C, a 18650 Li-ion battery with 1.2 Ah, in its pristine state at various discharge rates (0.5, 1.0, and 2.0 C), achieved by applying currents of 0.63, 1.25, and 2.5 A. Using Eq. (1), the evolution of AEGD is determined and shown in Figure 38. Figure 39 presents the AEGD value up to 2.5V, confirming the linear relationship between AEGD and I, given by $AEGD = 0.33I + 16.5$. The evolution of AEGD with SOD is displayed in Figure 40. Next, the evolution of AEGD for 0.5 and 2.5 C discharge rates is normalized with respect to the 1.0 C discharge rate using the calculated ratios $R_{0.5} = 1.01$, $R_{2.0} = 0.975$, and the results are plotted in Figure 41. The plot ascertains that AEGD value with SOD follows almost the same path for different currents. The curve-fitting equation characterizing the evolution of AEGD and SOD, incorporating the ratios, is provided in Eq. (A13).

$$SOD_x = (31.1 \times AEGD^{0.99} - 24.22 \times AEGD) \times R_x \qquad \text{(A13)}$$

Figures 37-41 show experimental results for Battery C, a 18650 Li-ion battery with 1200 mAh. Figure 37 shows discharge voltage measured for different discharge current rates (0.5, 1.0, and 2.0 C). Figure 38 shows the evolution of entropy generation with different discharge currents. Figure 39 shows the linear fit of AEGD to current at discharge voltage of 2.5 V. Figure 40 shows normalized evolution of AEGD for different discharge rates with percentage of SOD.

The findings from the results from the three different types of batteries confirm that the AEGD value maintains a linear relationship with the applied current, allowing for the development of a generalized linear curve-fit equation to predict AEGD values across various discharge rates or applied currents. Following the normalization approach, using the AEGD ratio at any of the discharge rates as a reference effectively aligns the AEGD evolution curves across different currents or discharge rates in the same path. By leveraging the established curve-fit equations relating to the AEGD value and its evolution with varying rates of discharge, a procedure is proposed and validated in the discussion associated with Example Set 1 above to determine the SOD of a discharging battery. We propose that the developed procedure can reliably and efficiently determine the SOD in real-time, enabling a precise assessment of the battery's remaining useful life.

Figures 42-44 relate to a procedure for determining RDT using the proposed approach. Figures 42-44 show discharge voltage with time**,** evolution of AEGD value, and evolution of percentage of SOD with AEGD, for two different discharge rates, respectively.

Figure 45 is a schematic representation of the battery test setup.

Figures 46 and 47 show validation of the proposed approach for variable applied currents. Figures 46 and 47 show evolution of AEGD for low to high (1.25 → 2.5 → 5.0 A) and high to low (5.0 → 2.5 → 1.25 A) currents, for Batteries A and B, respectively.

Figure 48 is a schematic representation of discharge voltage curve demonstrating the three zones. Region 2 represents 80 % of the total discharge duration.

Figures 49 and 50 show validation of the proposed procedure for a fluctuating discharge voltage data with figure 49 depicting measured discharge voltages under both fluctuating and stable conditions and figure 50 depicting calculated AEGD for both conditions using Eq. (1).

As that phrase is used herein, “entropic change” includes for example accumulated entropy generation until complete discharge.

As that phrase is used herein, “relationships between current and entropy change” includes those relationships expressed by the equation $AEGD = A(I_i) + B$ and the equation $R_x = \frac{AEGD_{1.0}}{AEGD_x}$.

As that phrase is used herein, “technique(s) relating entropic changes to discharge progress metrics” include both the application of equation 5 in practice and the use of techniques which produce equivalent results such as the use of reference tables or the use of charts expressing such relationships.

As that phrase is used herein, a “discharge progress metric” includes for example remaining discharge time and state of discharge.

ADDITIONAL EXAMPLES

The following additional examples relate to the examples above and align with particular implementations of those methods.

1. A method of evaluating battery status comprising:
    a. measuring a discharge current of an evaluation battery;
    b. measuring a temperature of the evaluation battery:
    c. measuring a voltage across the evaluation battery;
    d. estimating a value for an entropic change associated with a battery discharge based on the measuring of the discharge current of the evaluation battery, the measuring the temperature of the evaluation battery, and the measuring of the voltage across the evaluation battery; and
    e. estimating a value of a discharge progress metric by a technique relating entropic changes to discharge progress metrics;
    f. wherein the technique relating entropic changes to discharge progress metrics further utilizes a relationship between discharge current and entropy change.
2. The method of 1 wherein the discharge progress metric is an estimated state of discharge of the evaluation battery.
3. The method of 1 wherein the discharge progress metric is a remaining discharge time of the evaluation battery.
4. The method of 1 wherein the technique relating entropic changes to discharge progress metrics comprises application of pre-determined correlation variables.
5. The method of 1 wherein the technique relating entropic changes to discharge progress metrics comprises application of non-linear correlation between pre-determined correlation variables.
6. The method of 1 wherein the estimating of the value for the entropic change associated with the battery discharge comprises an integration of variables including the discharge current of the evaluation battery, the temperature of the evaluation battery, and the voltage across the evaluation battery.
7. The method of 1 wherein the estimating of the value for the entropic change associated with the battery discharge comprises an integration of the general expression

$$\int_0^{t_d} \frac{[V_o - V(t)]I}{T(t)} dt$$

wherein I is the discharge current of the evaluation battery, T(t) is the temperature of the evaluation battery, V_0 is the open circuit potential of the evaluation battery and V(t) is the voltage across the evaluation battery.

8. The method of 1 further comprising conducting at least two tests on the evaluation battery to determine both the relationship between discharge current and entropy change and a relationship between entropic changes and at least one discharge progress metric prior to the estimating of the value of the discharge progress metric.
9. The method of 1 further comprising conducting at least two tests on a correlation battery having a construction similar to the evaluation battery to determine both the relationship between discharge current and entropy change and a relationship between entropic changes and at least one discharge progress metric prior to the estimating of the value of the discharge progress metric.
10. The method of 1 further comprising conducting at least two tests on a correlation battery having a construction similar to the evaluation battery to determine both the relationship between discharge current and entropy change and a relationship between entropic changes and at least one discharge progress metric prior to the estimating of the value of the discharge progress metric, wherein the discharge current of the evaluation battery is a current different than a set of currents utilized for the at least two tests.
11. The method of 1 wherein the discharge current varies over time.
12. The method of 1 wherein the voltage varies over time.
13. The method of 1 wherein the temperature varies over time.
14. The method of 1 wherein the measuring of the discharge current begins at an unknown state of discharge of the evaluation battery.
15. The method of 1 wherein the estimating of the value of the discharge progress metric is carried out in a micro-controller.

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The above-described embodiments have a number of independently useful individual features that have particular utility when used in combination with one another including combinations of features from embodiments described separately. There are, of course, other alternate embodiments which are obvious from the foregoing descriptions, which are intended to be included within the scope of the present application.